SmartCrop API Reference Manual

Complete Technical Specification & API Documentation for Backend Endpoints, Browser Web APIs, and Integrated Services

Base URL: http://localhost:8000/api	**Framework:** FastAPI (Python 3.14)	**Database:** SQLite + SQLAlchemy ORM
Frontend: React + Vite + Tailwind CSS	**Authentication:** OTP / Token & Session Auth	**API Docs:** /docs (Swagger UI)

1. Backend REST API Endpoints

The backend provides high-performance, asynchronous RESTful endpoints powering farmer authentication, distress calculation, crop advisory, chatbot interaction, and agricultural monitoring.

1.1 Authentication & Access Control

Method	Endpoint	Description & Purpose	Payload / Parameters
POST	`/auth/request-otp`	Requests an OTP for farmer mobile login. Generates mock 4-digit OTP ('1234') in development mode.	`{"phone": "9876543210"}`
POST	`/auth/verify-otp`	Validates the 4-digit OTP for the specified phone and returns a bearer authentication token.	`{"phone": "9876543210", "otp": "1234"}`
POST	`/auth/officer-login`	Authenticates agricultural department officers with username/password credentials.	`{"username": "admin", "password": "123"}`

1.2 Farmer Profiles & Agricultural Records

Method	Endpoint	Description & Purpose	Payload / Parameters
GET	`/farmers/`	Lists registered farmers with optional pagination offset and limit query parameters.	`Params: skip=0, limit=100`
POST	`/farmers/`	Creates and registers a new farmer profile with regional, crop, soil, and financial data.	`{"name": "Ramesh", "phone": "9876543210", "district": "Pune", "crop": "Wheat", "soil_type": "Black", "loan_amount": 50000, "days_to_loan_due": 180}`
POST	`/farmers/{id}/records/`	Adds a climate & price record for farmer ID, automatically calculating the Distress Risk Score (0-100).	`{"rainfall_deviation_percent": -25.0, "mandi_price_drop_percent": 15.0}`

1.3 Advisory Engine & AI Voice/Chat Assistant

Method	Endpoint	Description & Purpose	Payload / Parameters
POST	`/advisory/`	Generates agronomic advice based on crop type, soil composition, rainfall variance, and preferred language.	`{"crop": "Wheat", "soil_type": "Black Soil", "rainfall_deviation_percent": -20, "language": "hi"}`
POST	`/chat`	AI conversational endpoint handling farmer questions about mandi rates, rain forecast, and pest control.	`{"message": "What is the mandi price for wheat?"}`

1.4 Officer Distress Dashboard & Automated Alerts

Method	Endpoint	Description & Purpose	Payload / Parameters
GET	`/dashboard-data/`	Aggregates total farmers, counts high-risk profiles (Distress Score > 60), and returns ranked risk cases.	`Response: {total_farmers, high_risk_count, high_risk_farmers: [...]}`
POST	`/alert/{farmer_id}`	Triggers an automated SMS distress alert to the farmer's registered phone and alerts regional officers.	`Path param: farmer_id (int)`

2. Browser Client-Side Web APIs

The frontend leverages native browser Web APIs to provide high-accessibility voice capabilities for farmers in regional languages and persistent state management.

API Name	Interface / Constructor	Implementation & Feature Usage	Supported Locales

Speech Recognition	`SpeechRecognition / webkitSpeechRecognition`	Powers real-time **Speech-to-Text (STT)** microphone input in FarmerChat.jsx.	`en-IN (English India)`
			`hi-IN (Hindi)`
			`or-IN (Odia)`

Speech Synthesis	`window.speechSynthesis` `SpeechSynthesisUtterance`	Provides **'Read Aloud' (TTS)** voice feedback for assistant responses with Indian-accent voices.	Matches active app language (English, Hindi, Odia)
Web Storage API	`window.localStorage`	Persists session tokens (token, officerToken) and recent phone/username history.	Client-side storage

3. External Services & Distress Formula

Google Translate Service (googletrans): Integrated into advisory_engine.py to deliver multi-language agricultural advice dynamically.

Distress Scoring Algorithm: Calculates farmer vulnerability (0–100) using weighted multi-factor normalization:

Factor	Weight	Normalization Metric	Distress Trigger Threshold
Rainfall Deficit	**40% (0.4)**	`min(max(-rainfall_dev, 0), 50) / 50.0`	Severe deficit below -20% to -50%
Mandi Price Drop	**40% (0.4)**	`min(max(price_drop, 0), 30) / 30.0`	Price drop exceeds 15% to 30%
Loan Due Proximity	**20% (0.2)**	`max(365 - days_to_loan, 0) / 365.0`	Due date approaching 0–30 days
Total Score	**100%**	**Risk = (0.4·Rain + 0.4·Price + 0.2·Loan) × 100**	**Score > 60: Flagged as High Risk**